H0 — decision summary

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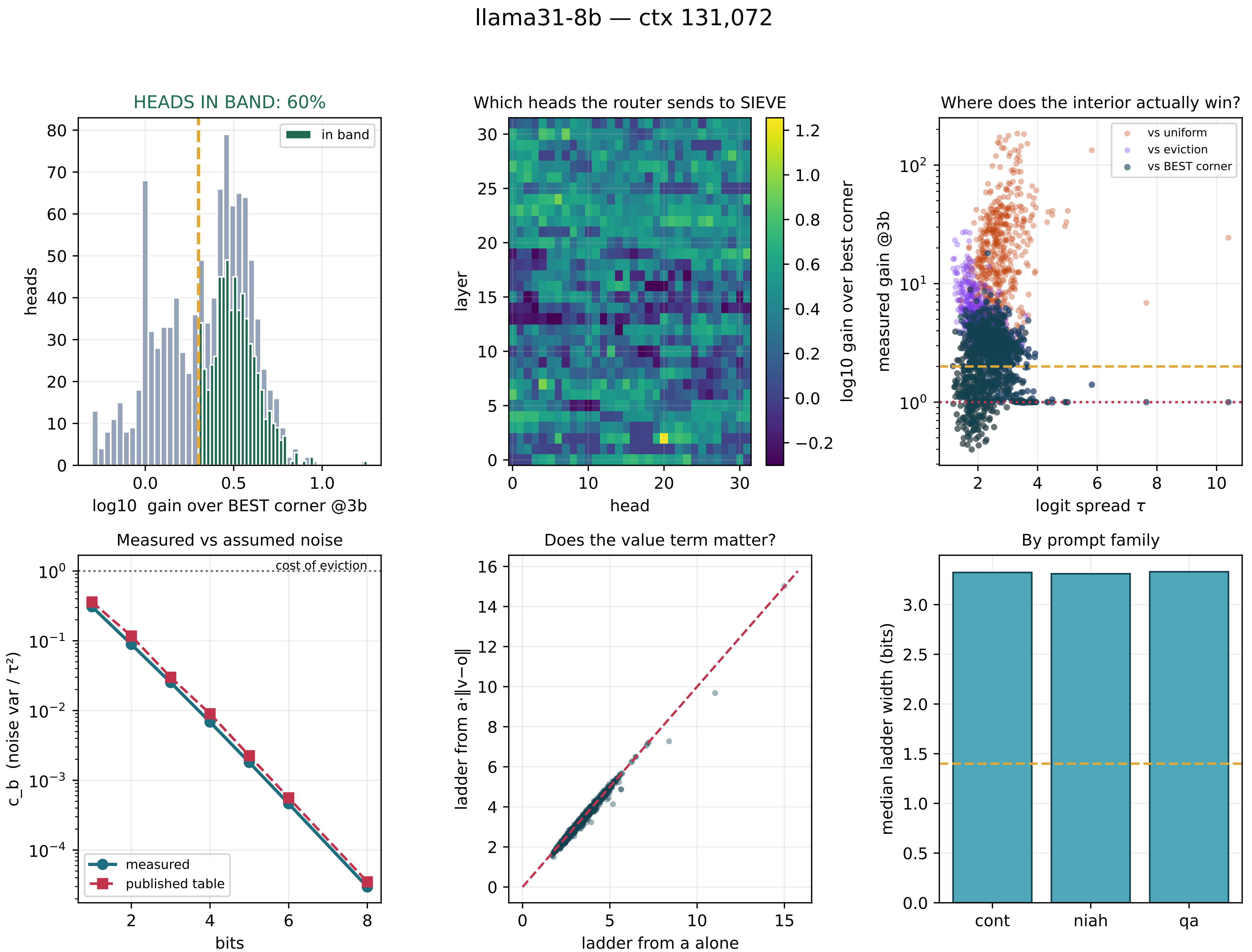
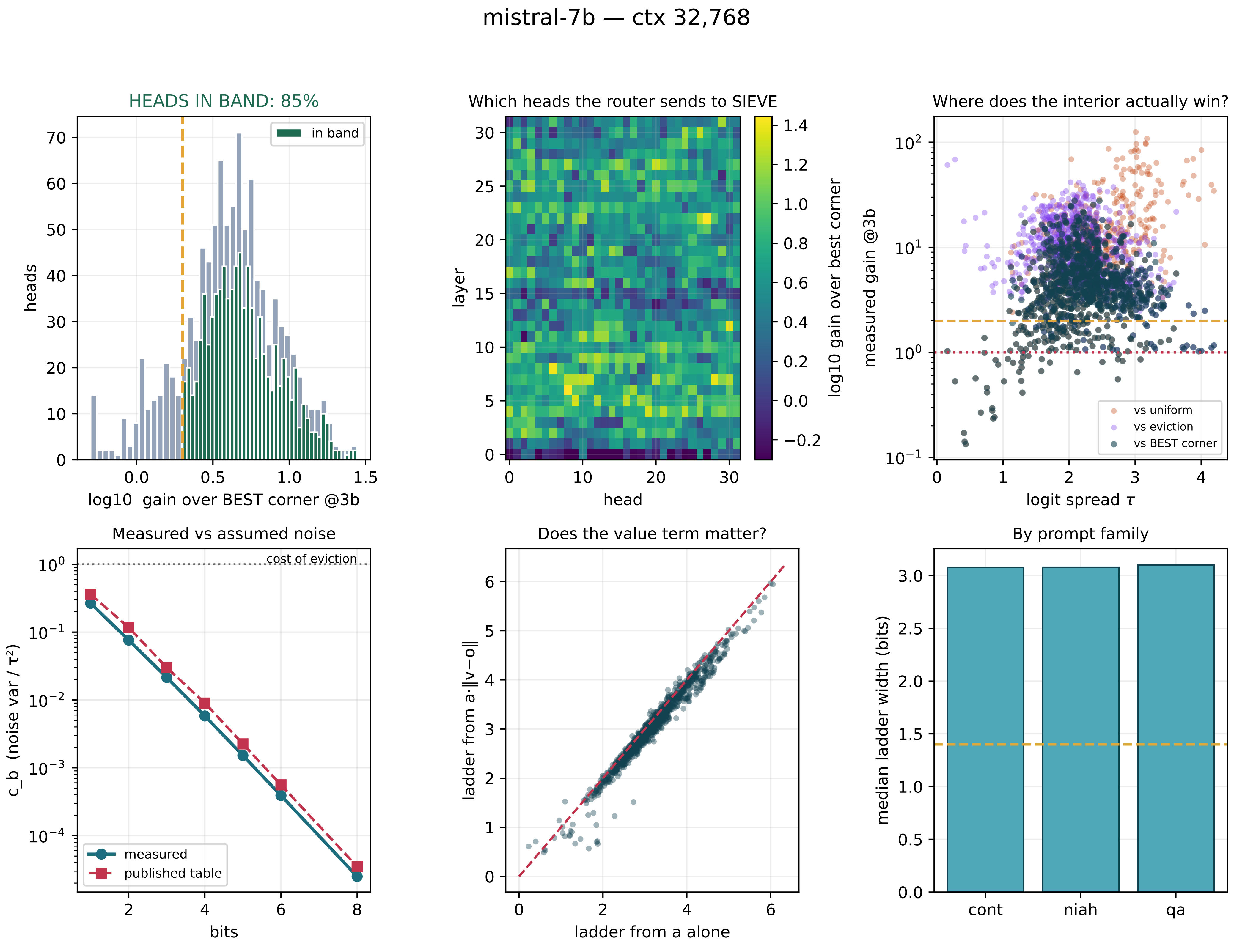
llama31-8b  (ctx 131,072,  1,024 heads)
    HEADS IN BAND   60.4%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     2.23x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   3.2x   p90   4.7x
    ladder width   median  3.32 b   IQR [2.78, 3.92]   100% of heads above 1.4
    tau            median  2.34   excl. sinks  2.34
    top-1 weight   median 0.0102   95% mass in 42456 tokens (35.2% of ctx)
    B=2b  vs uniform    4.7x   vs eviction    2.8x   vs BEST corner    2.3x   evicts 58%
    B=3b  vs uniform    4.9x   vs eviction    3.7x   vs BEST corner    2.5x   evicts 43%
    quantizer alpha @2b 0.941 (1.0 = no temperature distortion);  spearman@2b 0.530  [H2 answered]
    heads where eviction beats a 1-bit tier: 85%
    heads where eviction beats a 2-bit tier: 8%
    linearization check (predicted/measured gain @3b): 3.96  (1.0 = exact)
    VERDICT: GO — 60% of heads are in band (2.23x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test

mistral-7b  (ctx 32,768,  1,024 heads)
    HEADS IN BAND   84.7%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     4.15x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   4.7x   p90  11.0x
    ladder width   median  3.09 b   IQR [2.58, 3.61]   98% of heads above 1.4
    tau            median  2.21   excl. sinks  2.21
    top-1 weight   median 0.0051   95% mass in 11748 tokens (39.0% of ctx)
    B=2b  vs uniform    5.9x   vs eviction    3.8x   vs BEST corner    3.1x   evicts 52%
    B=3b  vs uniform    6.5x   vs eviction    7.5x   vs BEST corner    4.2x   evicts 35%
    quantizer alpha @2b 0.940 (1.0 = no temperature distortion);  spearman@2b 0.881  [H2 answered]
    heads where eviction beats a 1-bit tier: 74%
    heads where eviction beats a 2-bit tier: 1%
    linearization check (predicted/measured gain @3b): 2.31  (1.0 = exact)
    VERDICT: GO — 85% of heads are in band (4.15x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test

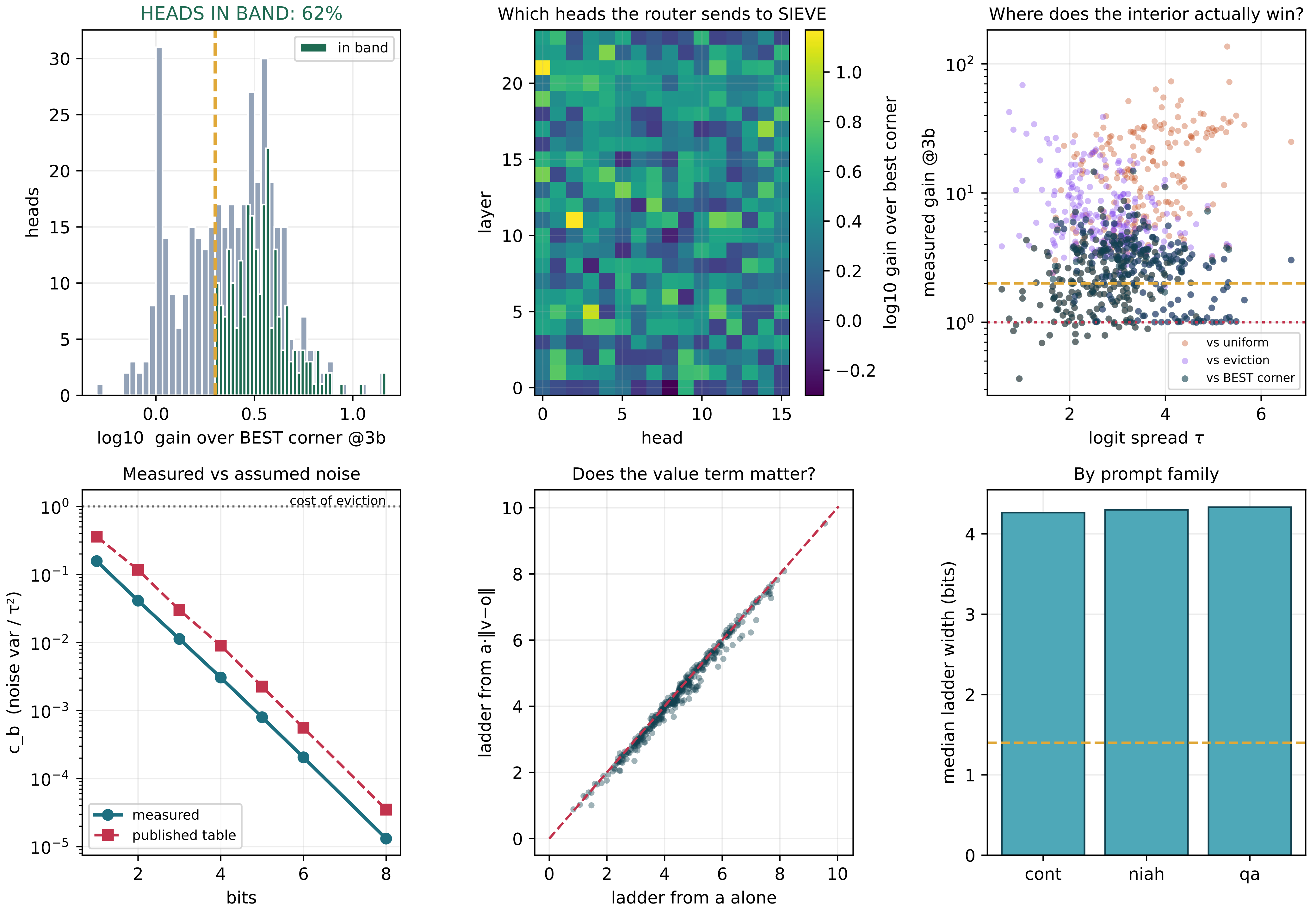
qwen15-moe-a2.7b  (ctx 32,768,  384 heads)
    HEADS IN BAND   62.0%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     2.38x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   3.3x   p90   5.3x
    ladder width   median  4.28 b   IQR [3.33, 5.27]   98% of heads above 1.4
    tau            median  3.07   excl. sinks  3.07
    top-1 weight   median 0.0055   95% mass in 11257 tokens (37.3% of ctx)
    B=2b  vs uniform    3.8x   vs eviction    3.1x   vs BEST corner    2.4x   evicts 57%
    B=3b  vs uniform    4.2x   vs eviction    4.5x   vs BEST corner    2.5x   evicts 43%
    quantizer alpha @2b 0.943 (1.0 = no temperature distortion);  spearman@2b 0.556  [H2 answered]
    heads where eviction beats a 1-bit tier: 80%
    heads where eviction beats a 2-bit tier: 3%
    linearization check (predicted/measured gain @3b): 3.18  (1.0 = exact)
    VERDICT: GO — 62% of heads are in band (2.38x routed). The interior owns a substantial share of the model. Proceed to the nested-coding test

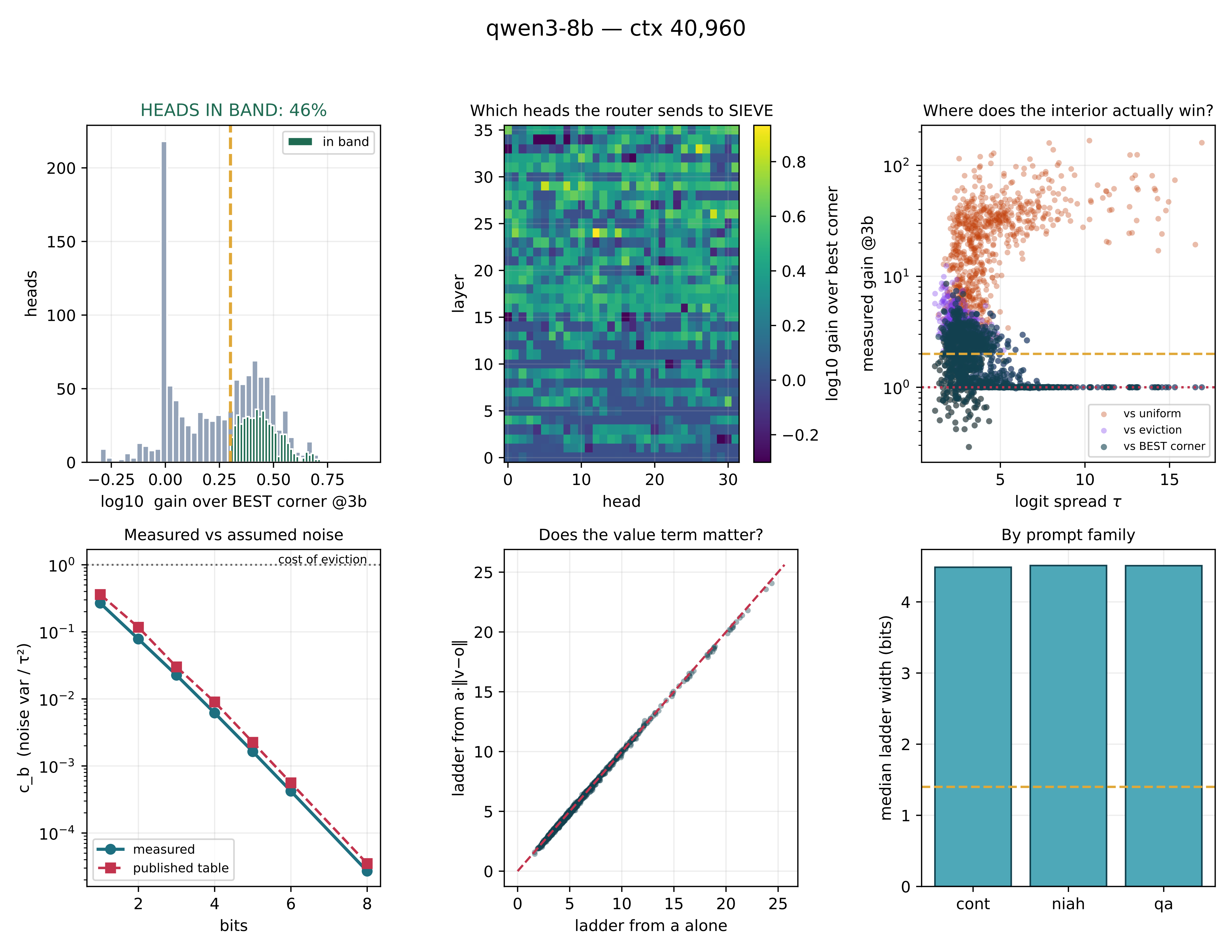
qwen3-8b  (ctx 40,960,  1,152 heads)
    HEADS IN BAND   45.9%  (gain over best corner >= 2.0x at 3 b/token)
    ROUTED GAIN     1.77x  (geometric mean of max(gain,1) over all heads)
    in-band heads  median   2.7x   p90   3.7x
    ladder width   median  4.50 b   IQR [3.56, 6.02]   100% of heads above 1.4
    tau            median  3.17   excl. sinks  3.17
    top-1 weight   median 0.0317   95% mass in 9514 tokens (25.2% of ctx)
```

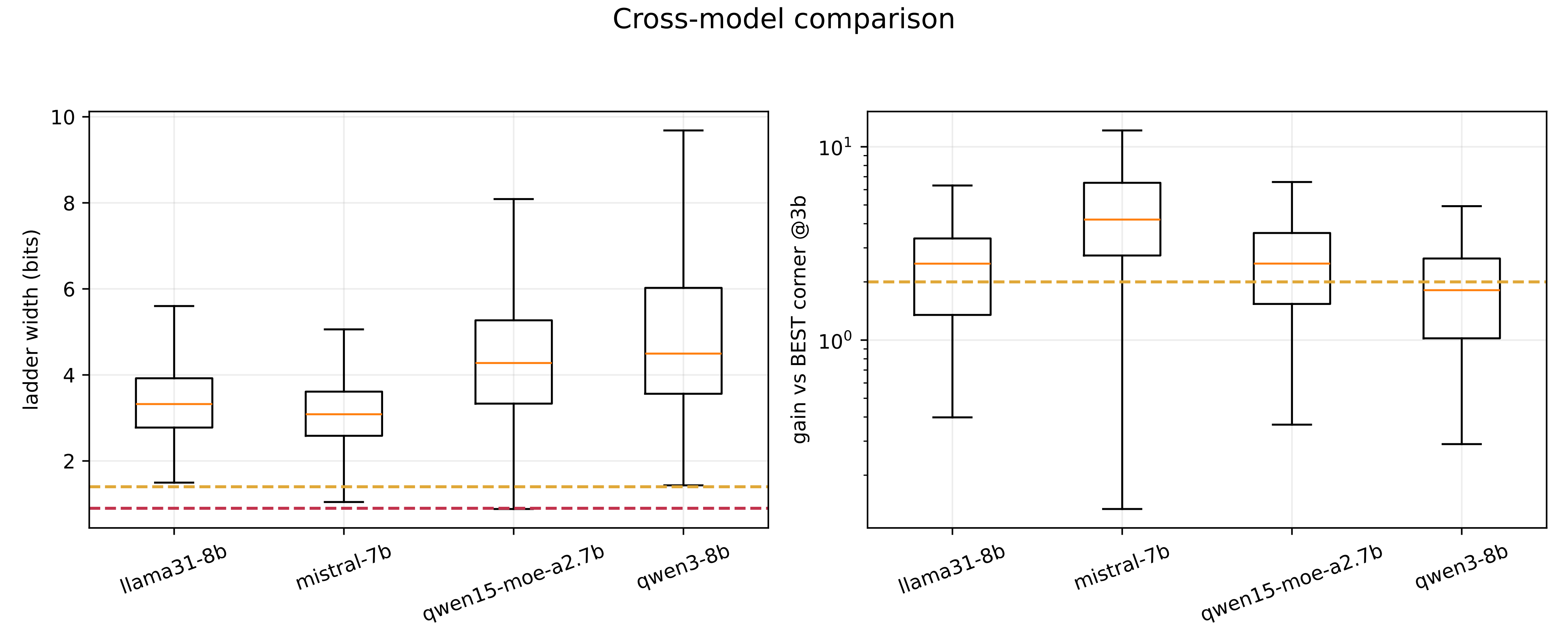

llama31-8b — ctx 131,072
HEADS IN BAND: 60%
in band
heads
log10 gain over BEST corner @3b
Which heads the router sends to SIEVE
layer
head
log10 gain over best corner
Where does the interior actually win?
vs uniform
vs eviction
vs BEST corner
measured gain @3b
logit spread τ
Measured vs assumed noise
cost of eviction
c_b (noise var / τ²)
bits
measured
published table
Does the value term matter?
ladder from a·‖v−o‖
ladder from a alone
By prompt family
median ladder width (bits)
cont
niah
qa


mistral-7b — ctx 32,768
HEADS IN BAND: 85%
in band
heads
log10 gain over BEST corner @3b
Which heads the router sends to SIEVE
layer
head
log10 gain over best corner
Where does the interior actually win?
measured gain @3b
logit spread τ
vs uniform
vs eviction
vs BEST corner
Measured vs assumed noise
cost of eviction
c_b (noise var / τ²)
bits
measured
published table
Does the value term matter?
ladder from a·‖v−o‖
ladder from a alone
By prompt family
median ladder width (bits)
cont
niah
qa

qwen15-moe-a2.7b — ctx 32,768




qwen3-8b — ctx 40,960
HEADS IN BAND: 46%
in band
heads
log10 gain over BEST corner @3b
Which heads the router sends to SIEVE
layer
head
log10 gain over best corner
Where does the interior actually win?
measured gain @3b
logit spread τ
vs uniform
vs eviction
vs BEST corner
Measured vs assumed noise
cost of eviction
c_b (noise var / τ²)
bits
measured
published table
Does the value term matter?
ladder from a·‖v−o‖
ladder from a alone
By prompt family
median ladder width (bits)
cont
niah
qa


Cross-model comparison
ladder width (bits)
10
8
6
4
2
llama31-8b
mistral-7b
qwen15-moe-a2.7b
qwen3-8b
gain vs BEST corner @3b
10^1
10^0
llama31-8b
mistral-7b
qwen15-moe-a2.7b
qwen3-8b